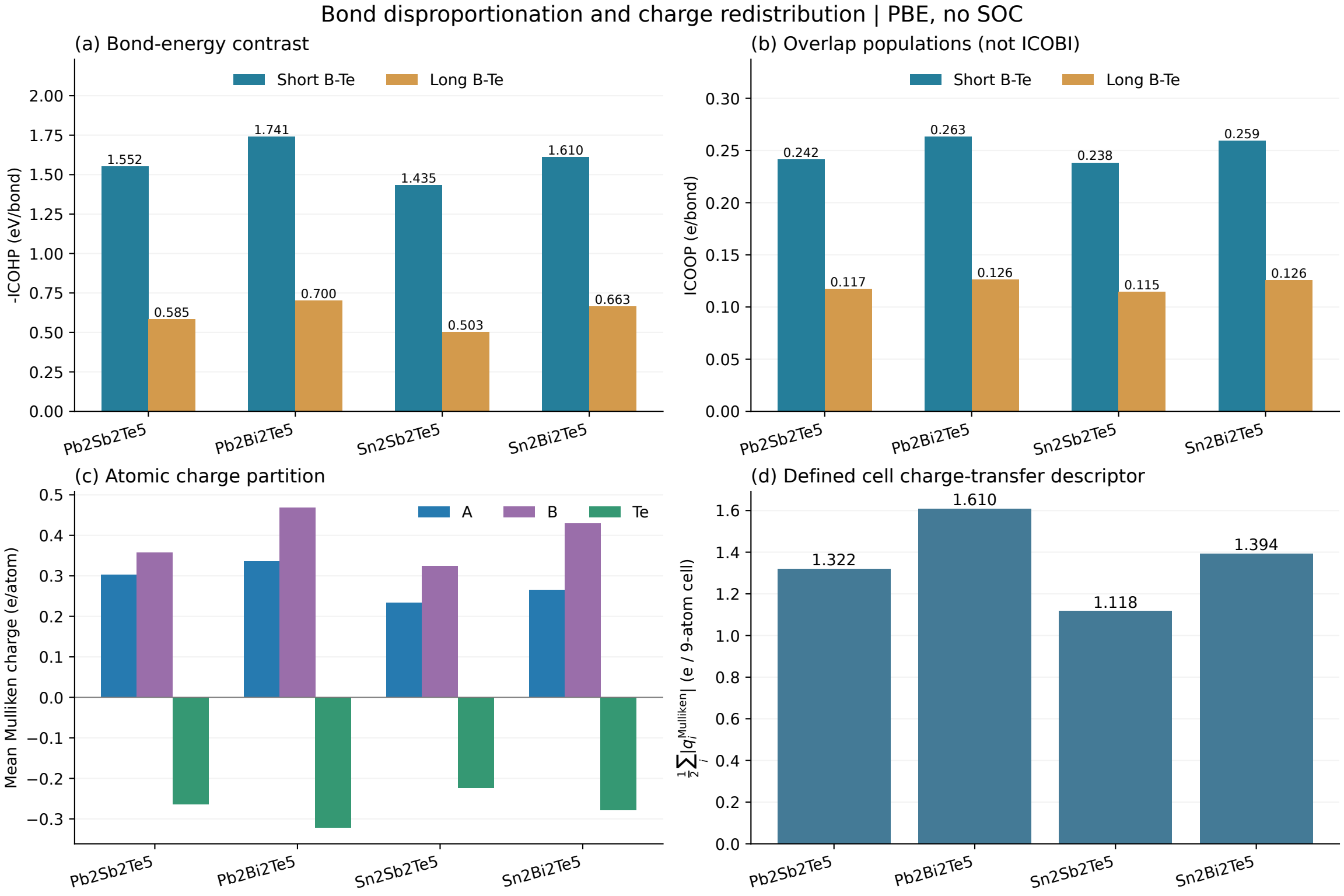
Bond disproportionation and charge redistribution | PBE, no SOC
(a) Bond-energy contrast
Short B-Te
Long B-Te
-ICOHP (eV/bond)
1.552
0.585
1.741
0.700
1.435
0.503
1.610
0.663
Pb2Sb2Te5
Pb2Bi2Te5
Sn2Sb2Te5
Sn2Bi2Te5
(b) Overlap populations (not ICOBI)
Short B-Te
Long B-Te
ICOOP (e/bond)
0.242
0.117
0.263
0.126
0.238
0.115
0.259
0.126
Pb2Sb2Te5
Pb2Bi2Te5
Sn2Sb2Te5
Sn2Bi2Te5
(c) Atomic charge partition
A
B
Te
Mean Mulliken charge (e/atom)
Pb2Sb2Te5
Pb2Bi2Te5
Sn2Sb2Te5
Sn2Bi2Te5
(d) Defined cell charge-transfer descriptor
$\frac{1}{2}\sum_i |q_i^{\mathrm{Mulliken}}|$ (e / 9-atom cell)
1.322
1.610
1.118
1.394
Pb2Sb2Te5
Pb2Bi2Te5
Sn2Sb2Te5
Sn2Bi2Te5